

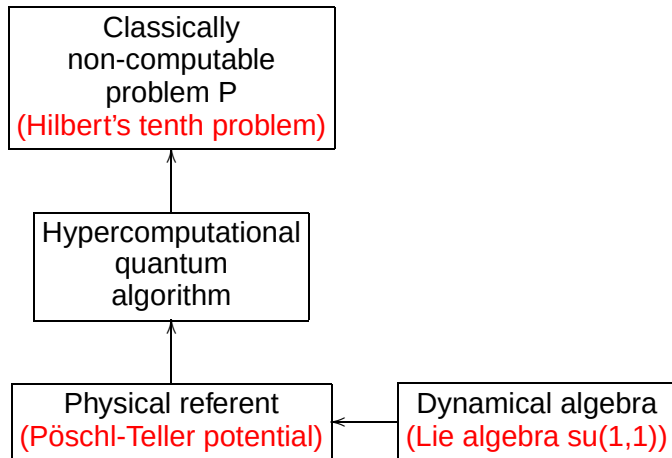
# Quantum algorithm of hypercomputation based on the Pöschl-Teller potential

Mario Vélez, Juan Ospina, and Andrés Sicard

Grupo de Lógica y Computación  
Universidad EAFIT, Medellín, Colombia

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# Key ideas



# Hypercomputation I

## Definition

A **hypercomputer** is any machine (theoretical or real) that compute functions or numbers, or more generally solve problems or carry out tasks, that cannot be computed or solved by a Turing machine (TM)

# Hypercomputation II

## Examples

- Oracle Turing Machine (Turing)
- Accelerating Turing machine (Copeland)
- Analog Recurrent Neural Network (Siegelmann and Stong)
- ⋮

## Implementation

The possibility of **real** construction of a hypercomputer is controversial and is still under analysis.

# Incomputable-(Turing Machine) problem

## Hilbert's tenth problem

Given a **Diophantine** equation

$$D(x_1, \dots, x_k) = 0 ,$$

we should build a procedure to determine whether or not this equation has a solution in  $\mathbb{N}$ .

**Classical solution (Matiyasevich, Davis, Robinson, Putnam)**

Hilbert's tenth  
problem

$\equiv$

Halting problem  
for Turing Machine

# Physical referent: Pöschl-Teller potential I

- Quantum system: particle with mass  $m$  trapped inside the Pöschl-Teller potential  $0 \leq x \leq \pi l$

$$V(x) = \frac{V_0}{2} \left( \frac{\lambda(\lambda-1)}{\cos^2 x/2l} + \frac{\kappa(\kappa-1)}{\sin^2 x/2l} \right)$$

$$H(x) = i^2 \frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) - \frac{V_0}{2}(\lambda + \kappa)^2 ,$$

- Computational basis and action of  $H$  on it:

$$\{ |\eta/2, n\rangle \mid n \in \mathbb{N} \} \quad \text{where } \eta = \lambda + \kappa + 1 ,$$
$$H |\eta/2, n\rangle = E_n |\eta/2, n\rangle , \quad \text{where } E_n = \hbar\omega n(n + \lambda + \kappa) .$$

# Physical referent: Pöschl-Teller potential II

- Dynamical Lie algebra  $\mathfrak{su}(1, 1)$  associated with PT:

$$[K_-, K_+] = K_3 \quad , \quad [K_{\pm}, K_3] = \mp 2K_{\pm} \quad .$$

- Infinite-dimensional irreducible representation for  $\mathfrak{su}(1, 1)$

$$K_+ |\eta/2, n\rangle = \sqrt{(n+1)(n+\eta)} |\eta/2, n+1\rangle \text{ (creation operator) } ,$$

$$K_- |\eta/2, n\rangle = \sqrt{n(n+\eta-1)} |\eta/2, n-1\rangle \text{ (annihilation operator) } ,$$

$$K_3 |\eta/2, n\rangle = (2n + \eta) |\eta/2, n\rangle \text{ (Cartan operator) } .$$

# Physical referent: Pöschl-Teller potential III

## ■ Number operator

$$N = (1/2)(K_3 - \eta) = \begin{pmatrix} 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ 0 & 0 & 2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} ,$$

$$N |\eta/2, n\rangle = n |\eta/2, n\rangle .$$

## ■ Barut-Girardello coherent states ( $K_- |\eta/2, z\rangle = z |\eta/2, z\rangle$ , with $z \in \mathbb{C}$ ):

$$|\eta/2, z\rangle = \frac{1}{\sqrt{\left\{ \Gamma(\eta) |z|^{-(\eta-1)} I_{\eta-1}(2|z|) \right\}}} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n! (\eta)_n}} |\eta/2, n\rangle .$$



# Algorithm's strategics I

## Codification à la Kieu

$$\begin{array}{ccc} D(x_1, \dots, x_k) = 0 & \xrightarrow{\text{codification}} & H_D = (D(N_1, \dots, N_k))^2 \\ \downarrow & & \downarrow \\ \text{Solution in } \mathbb{N} & \xrightarrow{\text{if and only if}} & H_D | \{n\} \rangle_0 = 0 \end{array}$$

Note: Continuous quantum computation

## New problem

To find the ground state  $|\{n\}\rangle_0$  of  $H_D$ .

# Algorithm's strategics II

## Solution à la Kieu: adiabatic quantum computation

$$H_A(t) = (1 - t/T)H_I + (t/T)H_D \text{ over } t \in [0, T]$$

$$H_I = \sum_{i=1}^k (K_{+i} - z_i^*) (K_{-i} - z_i), \quad |\psi(0)\rangle = \bigotimes_{i=1}^k |z_i\rangle .$$

# Hypercomputational Quantum Algorithm I

- 1 Construct a physical process subject to  $H_A(t)$  over the time interval  $[0, T]$ , for some **finite** time  $T$ .
- 2 Measure through the time-dependent Schrödinger equation  $i\partial_t |\psi(t)\rangle = H_A(t) |\psi(t)\rangle$ , for  $t \in [0, T]$  the maximum probability

$$P_{\max}(T) = \max_{(n_1, \dots, n_k) \in \mathbb{N}^k} |\langle \psi(T) | n_1, \dots, n_k \rangle|^2 = |\langle \psi(T) | \{n\} \rangle_0|^2.$$

- 3 If  $P_{\max}(T) \leq 1/2$ , increase  $T$  and repeat all the steps above.
- 4 If  $P_{\max}(T) > 1/2$  (**halting criterion**) then  $|\{n\}\rangle_0$  is the ground state of  $H_D$  (assuming no degeneracy).
- 5  $D(x_1, \dots, x_k) = 0$  has a solution in  $\mathbb{N}$ , **if and only if**,  $H_D |\{n\}\rangle_0 = 0$ .

# Conclusions

- *“Once upon on time, back in the golden age of the recursive function theory, computability was an absolute”<sup>1</sup>*
- The implementation of a hypercomputer is an open problem.
- Adiabatic continuos quantum computation  $\neq$  “standard” quantum computation.

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<sup>1</sup>Richard Sylvan and Jack Copeland. Computability is logic-relative. In Graham Priest and Dominic Hyde, editors, Sociative logics and their applications: essays by the late Richard Sylvan. London: Ashgate Publishing Company, 2000, p. 189.

# Further reading I



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# Further reading II



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Mario Elkin Vélez  
*email:* [mvelez@eafit.edu.co](mailto:mvelez@eafit.edu.co)